



Summer Training Project

Project: Retail Customer Purchase Behavior
& Sales Performance

Group Code: S26-B3-Data A-G7

Project Number: 1

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1.Introduction

The company has collected two years of online retail transaction data (2009–2011). However, despite having a large amount of sales data, it lacks clear insights into its business performance. The company does not fully understand which products generate the highest revenue, who its most valuable customers are, or how sales vary across different countries and times.

Without analyzing this data, important business decisions—such as inventory planning, marketing campaigns, and customer relationship strategies—are often based on assumptions rather than evidence. This can lead to missed sales opportunities, inefficient resource allocation, and reduced business performance.

2.Business Scenario

An online retail company aims to improve its business performance by making data-driven decisions. The company has accumulated two years of transactional sales data but lacks a clear understanding of customer purchasing behavior, product performance, and sales trends across different countries and time periods.

Management wants to identify the products that contribute the most to revenue, recognize their most valuable customers, understand seasonal demand patterns, and evaluate sales performance across international markets. These insights will support strategic decisions related to inventory management, marketing campaigns, customer retention, and future business planning.

As a data analyst, the objective of this project is to transform raw sales data into meaningful business insights through data exploration, cleaning, visualization, and analytical techniques. The outcome is intended to help management optimize operations and improve profitability for the next business quarter.

3.Dataset Description

This project uses the **Online Retail II** dataset obtained from the **UCI Machine Learning Repository**. The dataset contains transactional records from a UK-based online retail company that primarily sells unique gift items to customers across multiple countries.

The dataset is provided as an Excel workbook containing **two worksheets**, each representing a different period of sales transactions:

Worksheet	Time Period
Year 2009–2010	December 1, 2009 – December 9, 2010
Year 2010–2011	December 1, 2010 – December 9, 2011

Since both worksheets have the same structure and attributes, they were merged into a single dataset before the analysis. Combining the two worksheets provides a complete view of approximately two years (24 months) of sales activity, allowing more accurate trend analysis, customer segmentation, and business insights.

Each row in the dataset represents a single product within a customer invoice. Therefore, one invoice may contain multiple rows if multiple products were purchased in the same transaction.

The dataset includes the following attributes:

Attribute	Description
Invoice	Unique invoice number for each transaction. Cancelled invoices usually begin with the letter "C".
StockCode	Unique identifier assigned to each product.
Description	Product name or description.
Quantity	Number of units purchased. Negative values usually indicate returned or cancelled items.
InvoiceDate	Date and time when the transaction was recorded.
Price	Unit price of the purchased product.
Customer ID	Unique identifier assigned to each customer. Some records have missing customer IDs.
Country	Country where the customer is located.

The dataset is suitable for performing sales analysis, customer behavior analysis, revenue analysis, customer segmentation using the RFM model, and identifying sales trends across different products, countries, and time periods.

4.Solution Steps

Python

Load data

2 Data Frames (each sheet in a DataFrame)

Exploration

1. Show top 10 rows of data
2. Shape
3. Columns name
4. Types of columns
5. Number of total transactions
6. Number of unique customers
7. Number of unique products
8. Number of countries
9. Date range
10. Name of unique countries
11. Description for new/working columns
12. Top 20 rows where Price = 0
13. Show new/working columns
14. Missing values
15. Duplicates
16. Outliers
17. Inconsistence (Negative Quantity and Price)
18. Identify canceled transactions

Cleaning

1. Remove duplicates
2. Fill Description by "Unknown"
3. Fill Description by Description has same StockCode
4. Handle CustomerID
5. Fill Price = 0 by price of the same StockCode and Description
6. Drop Price with null
7. Drop inconsistent Quantity
8. Drop inconsistent Price
9. Handle positive quantity in canceled transactions
10. Drop inconsistent Price
11. Standardizing fields
12. Separate DataFrame from canceled transactions

Integration

four Data Frame Integration Give the two Data Frames:

1. Uncanceled Transactions (df)
2. Canceled Transactions (df_canceled)

CSV

Convert two Data Frames to two CSV files:

- 1.NTI_Cleaned_Final.csv
- 2.NTI_Cancelled_Clean.csv

Check

-Check all data in NTI_Cleaned_Final.csv

1. No zero price
2. No negative values
3. No null

4. No duplicates

5. No canceled transaction

-Check all data in NTI_Canceled_Cleaned_Final.csv

1. No zero price

2. No positive values

3. No null

4. No duplicates

5. No uncanceled transaction

SQL

Business and Diagnostic Analysis Questions

Setup in SQL server

	Column Name	Data Type
►	Invoice	nvarchar(50)
	StockCode	nvarchar(50)
	Description	nvarchar(150)
	Quantity	int
	InvoiceDate	datetime2(7)
	Price	decimal(10, 5)
	Customer_ID	float
	Country	nvarchar(50)

First, make Database with any name and make 2 tables cleaned and canceled

And this is the valid design that we used

Product Analysis

Q1. What are the top 10 products by revenue?

Purpose

Identify the products that generate the highest total revenue to understand which items contribute most to overall sales.

Expected Insight

- Top-selling products by revenue and not cancelled.
- Useful for inventory planning and marketing campaigns.

Q2. Which products generate the least revenue?

Purpose

Identify products with poor sales performance.

Expected Insight

- Low-performing products.
- Candidates for promotions or discontinuation.

Q3. Which products generate high revenue despite low sales quantity?

Purpose

Find premium or expensive products that earn high revenue even with low sales volume.

Expected Insight

- High-value products.
- Luxury or premium items.

Q4. Which products have high sales quantity but low revenue?

Purpose

Identify products sold in large quantities but generating relatively low revenue.

Expected Insight

- Low-priced products.
- Frequently purchased items.

Q5. Which products are canceled most frequently and what percentage do they represent?

Purpose

Analyze product cancellations to identify problematic items.

Expected Insight

- Frequently canceled products.
- Possible issues with quality, availability, or customer satisfaction.

Q6. Which pairs of products are most purchased together?

Purpose

Discover products frequently bought together.

Expected Insight

- Product bundles.

- Cross-selling opportunities.
- Recommendation systems.

Customer Analysis

Q7. Which customers order most frequently, and what is their average order value?

Purpose

Identify loyal customers and evaluate their purchasing behavior.

Expected Insight

- Most active customers.
- Average spending per order.

Q8. When did each customer place their last order?

Purpose

Determine the most recent purchase date for every customer.

Expected Insight

- Active customers.
- Inactive customers.
- Customer retention analysis.

Q9. Segment customers using RFM Analysis (Recency, Frequency, Monetary)?

Purpose

Classify customers into business segments based on purchase behavior.

Customer Segments

- High-Value Customers
- At-Risk Customers
- New Customers

Expected Insight

- Identify loyal customers.

- Detect customers likely to churn.
- Recognize newly acquired customers.

Time Analysis

Q10. How does monthly revenue change over the two years?

Purpose

Analyze revenue trends over time.

Expected Insight

- Monthly sales trend.
- Seasonal demand.
- Growth or decline in revenue.
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Q11. Which day of the week generates the highest revenue?

Purpose

Identify the best-performing weekday.

Expected Insight

- Peak shopping days.
- Better scheduling for promotions and staffing

Country Analysis

Q12. Which countries generate the most revenue, and how concentrated is it?

Purpose

Determine revenue contribution by country.

Expected Insight

- Top revenue-generating countries.
- Revenue concentration by market.

Example Finding

The United Kingdom contributes approximately 86% of the total revenue.

Q13. Which country has the highest average quantity per order ?

Purpose

Compare purchasing behavior across countries.

Expected Insight

- Countries with larger average order sizes.
- Difference in market demand.

Revenue Analysis

Q14. What is the net revenue after removing discounts?

Purpose

Calculate the actual revenue after excluding discount transactions

Expected Insight

- Gross Revenue
- Total Discounts
- Net Revenue

Excel

Get Answers for All Questions (#14) Above and make dashboards

5.Visualization (Power BI)

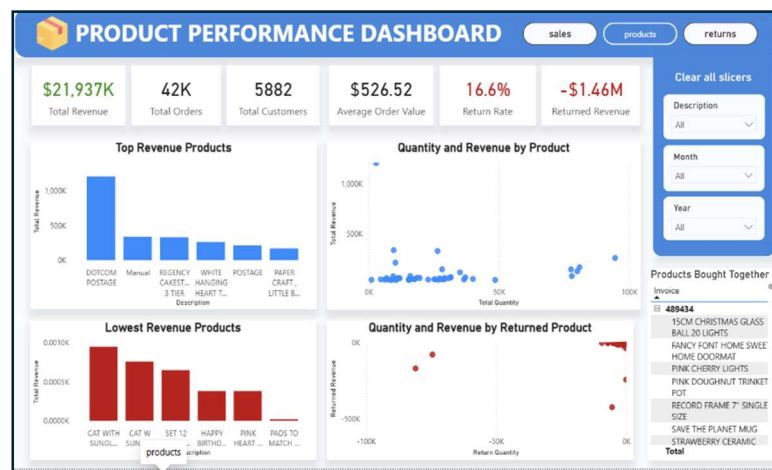
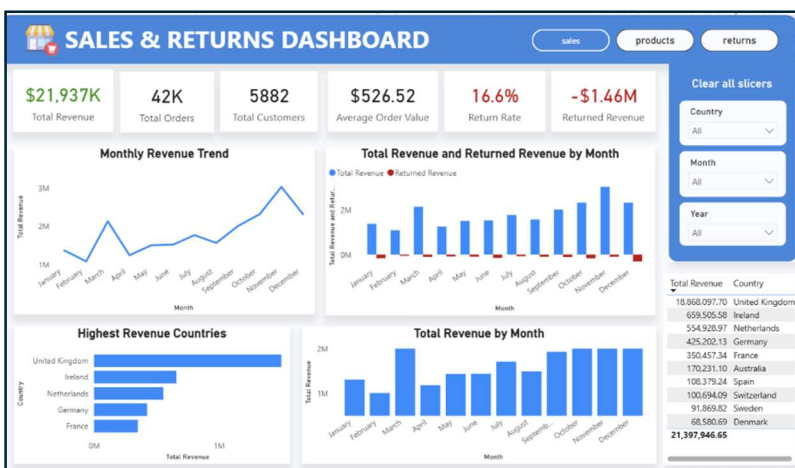
1-Import Data

2-Data Modeling

3-Create Measures

4-Create Charts and tables

5-Create Interactive dashboard with 3 Pages (sales / products / returns)



6. Insights and recommendations

1. Product Performance

Insight

The analysis shows that **high revenue is not always driven by high sales volume**. Some products generate significant revenue despite selling fewer units, meaning profitability is more important than quantity sold. In addition, products with high cancellation rates indicate potential issues with product quality, customer expectations, or the ordering process, leading to avoidable revenue loss.

Recommendation

- Prioritize high-profit products in marketing campaigns.
 - Investigate the root causes of cancellations and improve product quality, descriptions, or fulfillment processes before increasing inventory.
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2. Monthly Revenue Trend

Insight

Revenue fluctuates throughout the year and is consistently higher on weekdays than weekends. This indicates that **customer purchasing behavior is seasonal and time-dependent**, meaning demand cannot be managed using a fixed sales strategy.

Recommendation

- Increase inventory and marketing activities before peak sales periods.
 - Launch weekend promotions to stimulate demand during lower-performing days.
-

3. Customer Behavior (RFM Analysis)

Insight

RFM analysis reveals that **customers contribute different levels of business value**. High-value customers generate a significant share of revenue, while at-risk customers represent a potential loss of future sales if no retention strategy is implemented.

Recommendation

- Develop targeted retention campaigns for at-risk customers.
 - Reward high-value customers with loyalty programs and exclusive offers.
 - Encourage new customers to make repeat purchases through personalized follow-up promotions.
-

4. Country Performance

Insight

The United Kingdom contributes the largest share of total revenue, indicating that **the business relies heavily on a single market**. This concentration increases business risk because any market changes in the UK could significantly impact overall revenue. Additionally, purchasing behavior varies across countries, suggesting that customers in different markets have different buying patterns.

Recommendation

- Expand marketing efforts in other high-potential countries to reduce dependence on the UK.
 - Customize promotions and product strategies based on customer behavior in each market.
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5. Product Association

Insight

Frequently purchased product combinations indicate that customers follow **consistent buying patterns**. This creates an opportunity to increase the average order value by encouraging customers to purchase complementary products together.

Recommendation

- Create product bundles for commonly purchased items.
 - Implement cross-selling recommendations during the checkout process.
-

Final Business Insight (Summary)

The analysis indicates that **business performance is driven by a combination of high-profit products, high-value customers, and strong performance in the UK market rather than by sales volume alone. At the same time, high product cancellation rates and heavy reliance on a single market expose the business to revenue risks. Customer purchasing behavior also varies by time and location, highlighting the importance of data-driven inventory, marketing, and customer retention strategies instead of applying a one-size-fits-all approach.**

Summary of Recommendations

1. **Focus on product profitability**, not just sales volume, and reduce cancellations by addressing their root causes.
2. **Use RFM segmentation** to retain high-value customers and re-engage at-risk customers with personalized campaigns.
3. **Align inventory and marketing with seasonal demand**, while using weekend promotions to improve low-performing periods.
4. **Reduce reliance on the UK market** by expanding into other promising countries and tailoring strategies for each market.
5. **Leverage product associations** through bundles and cross-selling to increase the average order value.